

TELCO CUSTOMER CHURN ANALYSIS REPORT

Insights & Recommendations to Reduce Customer Attrition

Data Source: Kaggle

26.5%

Overall Churn Rate

Seniors

Highest Risk Group

Month-to-Month

Riskiest Contract

E-Check

Riskiest Payment

01 SUMMARY

This report presents a comprehensive analysis of customer churn behaviour derived from the Telco Customer Churn dataset. The analysis identifies key drivers behind customer attrition and provides actionable recommendations to help reduce the churn rate from its current level of 26.5%.

**KEY
STAT**

26.54% of all customers have churned which is roughly 1 in 4 customers is leaving the service.

The analysis reveals that churn is driven by four core factors:

- Short customer tenure (1–2 months) with no early engagement hooks
- Month-to-month contracts offering no long-term commitment incentive
- Absence of value-added services like Online Security and Tech Support
- Electronic check payment method correlated with lower commitment

02 KEY FINDINGS

→ **Gender has no effect on churn**

Churn rates are nearly identical across male and female customers. Gender is not a predictive factor and should not be used to target retention campaigns — resources are better spent elsewhere.

▲ **Senior citizens churn at a disproportionately high rate** **HIGH RISK**

A significantly greater percentage of senior citizens have churned compared to non-senior customers. This demographic may struggle with complex digital products and require dedicated, simplified support and tailored plan offerings.

! **New customers (1–2 months) are the most vulnerable** **CRITICAL**

Customers who have been with the service for only 1–2 months churn at the highest rate. Long-tenured customers, on the other hand, are highly loyal. This points to a clear gap in the onboarding and early engagement experience.

✓ **Long-term contracts dramatically retain customers** **POSITIVE**

Customers on 1- or 2-year contracts are significantly less likely to churn compared to month-to-month subscribers. The contractual commitment creates stickiness and long-term value for both the customer and the business.

🔑 **Missing add-on services are a churn driver**

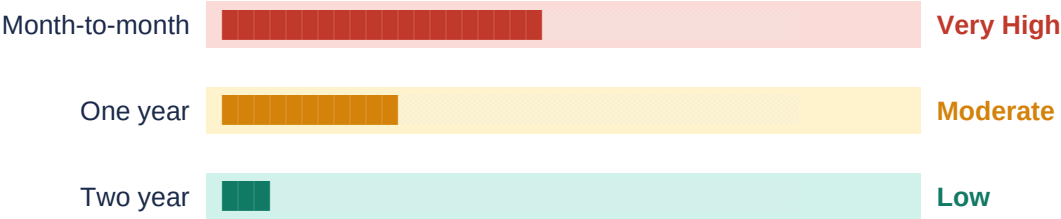
Customers without OnlineSecurity, TechSupport, OnlineBackup, and DeviceProtection show noticeably higher churn. Customers who engage with more services feel greater perceived value and are more likely to stay.

\$ Electronic check payment is a strong churn signal HIGH RISK

Customers paying by electronic check churn at the highest rate among all payment methods. This may reflect lower product commitment, a less digitally-engaged segment, or friction in the billing process itself.

03 **CHURN RISK BY CONTRACT TYPE**

The chart below illustrates the relative churn risk across contract types. Month-to-month subscribers represent the highest-risk group by a substantial margin.



INSIGHT

Customers on 2-year contracts are approximately 7x less likely to churn than month-to-month customers — converting even a small percentage of flexible subscribers would dramatically move the needle.

04 **SERVICE ADOPTION & CHURN**

Analysis across nine service dimensions reveals a consistent pattern: customers who lack security, support, and backup services churn at markedly higher rates.

Online Security Customers without this service churn noticeably more — perceived vulnerability.	Tech Support No support access leads to frustration and abandonment.	Online Backup Missing backup services correlate with higher churn rates.
Device Protection Customers without device cover feel undervalued.	Streaming TV Non-subscribers show slightly elevated churn.	Internet Service (DSL) DSL users churn less than Fiber Optic subscribers.

05 RECOMMENDATIONS

Based on the analysis, the following five strategies are recommended to reduce churn and improve long-term customer retention:

01

Structured Onboarding for New Customers (Months 1–3)

Since churn peaks in the first 1–2 months, launch a guided onboarding program, welcome calls, setup assistance, and a loyalty reward at the 3-month milestone, to anchor early users before they consider leaving.

02

Incentivize Upgrades from Month-to-Month to Annual Plans

Offer meaningful discounts (e.g., 1–2 months free) to encourage contract upgrades. Annual subscribers churn far less: converting even 15–20% of month-to-month customers would significantly reduce the overall churn rate.

03

Create a Senior Citizen Support Program

Assign dedicated support agents for senior customers, offer simplified onboarding materials, and consider a tailored plan with assisted setup. Proactive check-ins and human touchpoints can significantly reduce churn in this demographic.

04

Bundle Security and Support Services

Offer Online Security, Tech Support, Online Backup, and Device Protection as a discounted bundle or include them in base plans. Customers with more engaged service portfolios stay longer and generate more lifetime value.

05

Migrate Electronic Check Users to Auto-Pay

Offer a small bill credit or discount to customers who switch from electronic check to automatic payment. Automated billing reduces friction, is linked to higher retention, and is a low-cost intervention with measurable impact.

06 CONCLUSION

The current churn rate of 26.5% represents a significant business risk, but the data points to clear, addressable root causes. The most impactful interventions are early-tenure engagement, contract migration incentives, and bundled service adoption.

Gender is not a useful segmentation dimension for churn prevention. Instead, focus retention resources on: new customers (months 1–3), senior citizens, month-to-month subscribers, and electronic check users: these four groups account for the bulk of observed churn.

**PRIORITY
ACTION**

Begin with an onboarding improvement program, it targets the highest-churn window and requires no product changes, only process investment.